However, it speeds up the attainment of equilibrium by providing a new pathway having a lower activation energy.

For example, in the synthesis of $NH_3$ by the Haber's process iron is used as a catalyst. Similarly, in the contact process of manufacturing $SO_3$, platinum or $V_2O_5$ is used as a catalyst.

### 8.8.5 Effect of inert gas

When an inert gas (i.e, a gas which does not react with any other species involved in equilibrium) is added to an equilibrium system at constant volume, the total number of moles of gases present in the container increases, that is, the total pressure of gases increases. the partial pressure of the reactants and the products are unchanged. Hence at constant volume, addition of inert gas has no effect on equilibrium.

Table 8.3: Effect of concentration, pressure, temperature, catalyst and Inert gas on equilibrium

| Condition | Stress | Direction in which equilibrium shifts |
|---|---|---|
| **Concentration** | Addition of reactants (increase in reactant concentration) | Forward reaction |
| | Removal of products (decrease in product concentration) | |
| | Addition of products (increase in product concentration) | Reverse reaction |
| | Removal of reactants (decrease in reactant concentration) | |
| **Pressure** | Increase of pressure (Decrease in volume) | Reaction that favours fewer moles of the gaseous molecules |
| | Decrease of pressure (Increase in volume) | Reaction that favours more moles of the gaseous molecules |
| **Temperature** (Alters equilibrium constants) | Increase (High T) | Towards endothermic reaction |
| | decrease (Low T) | Towards exothermic reaction |
| **Catalyst** (Speeds up the attainment of equilibrium) | Addition of catalyst | No effect |
| **Inert gas** | Addition of inert gas at constant volume | No effect |



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